

Coding Assignment #2

August 4, 2025

Directions:

1. You will be penalized for not following each of the directions exactly as written below. If you aren't sure about what something means, ask the professor!
2. Put all your work in a single Haskell file.
3. Submit your work (before the deadline) to Canvas.
1. (Call your function **fenum**) Using only techniques from chapters 1-2, write a function that has the type `Enum a => a -> a -> a`; You may not use explicit type declaration.
2. What is the type of this function (answer first without using either Haskell or the Web)?

```
f a b = if a < b then [a..b] else [b..a]
```

3. Why doesn't `f` work with the arguments `(4, 5)` and `(8, 10)`? Explain your answer using the terms from Chapter 2.
4. (Call your function **toBinary**) Using only techniques from chapters 1-2, use explicit type declaration to write a function that takes an `Int` and converts it into a list of 1's and 0's. The **most significant** bits should be at the beginning of the list. You should assume that the argument to the function always has exactly 8-bits.
5. (Call your function **bitwiseOr**) Using only the techniques from chapters 1-2, use explicit type declaration to write a function that takes two `Int` arguments and computes their bit-wise or.
6. Your function must be called **calculateMatch**. Using only techniques from chapters 1-3, write a function that takes in two 3-tuples, call them (x_1, x_2, x_3) and (y_1, y_2, y_3) . Your program should output a String of 3 characters: for each i such that $x_i = y_i$, there should be one '*' character in the output String. If there are any i, j such that $y_i \neq x_i$ and $y_j \neq x_j$

but $y_i = x_j$, then there should be one corresponding '+' character in the output String. In order to ensure the response is always 3 characters long, fill in the rest with the character '-'. The output String must always put '*'s first, then '+'s, and lastly '-'. **Example:** `calculateMatch (3, 3, 5) (5, 3, 2)` should produce `"*+-"` (the 3s match in position and value ('*'), the 5s match in value but not position ('+'), and the other 3 and the 2 don't match at all ('-')). **Hints:** (1) You will find the function `threetuptolist (x, y, z) = [x, y, z]` useful (2) Note that, for 3-tuples being matched, it is impossible for there to be two instances of '+' arising from the same value. In other words, a 2, for example, can't appear twice in both lists and each time be responsible for producing a '+' in the output. This simplifies the computation.